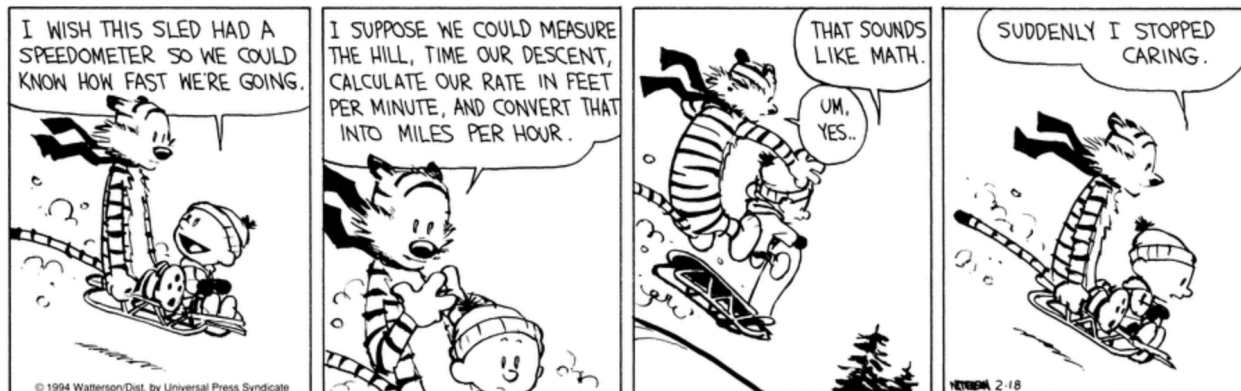


Course Information			Instructor Information	
Meeting Time:	MWF	3pm – 3:50pm	Name:	Zachary Porat (he/him)
	Th	1pm – 2:20pm	Email:	z.porat@bucknell.edu
Location:	MWF	RCHM 018	Office:	Olin 375
	Th	OLIN 268	Office Hours:	W 10am - 12pm
Website:	Moodle			Th 2:30pm - 3:30pm



Course Overview

Description: The goal of this course is to gain an understanding of how to quantify rates of change. We will examine methods for modeling systems that change over time, and learn how to use these models to predict future change. Specific topics of the course include theoretical and practical aspects of limits and derivatives; the calculus of exponential, logarithmic, and trigonometric functions; plane analytic geometry; an introduction to integral calculus; the fundamental theorem of calculus; various applications to medicine, biology, engineering, finance and more!

Text: The primary textbook for this course will be *Calculus, Early Transcendentals*, by Jon Rogawski, Colin Adams, and Robert Franzosa.

Learning Objectives: Below are several learning objectives for the semester.

- Develop effective problem-solving strategies by working with complex problems stemming from mathematical models of real-world situations.
- Understand and apply fundamental principles and theorems of differential calculus.
- Explain the connections between tangent lines, rates of change, and derivatives.
- Apply derivatives to a variety of practical problems including optimization, related rates, and the analysis of function behavior and graphs.

Office Hours / Drop-In Hours: I will hold office hours/drop-in hours on Wednesdays from 10am - 12pm and on Thursdays from 2:30pm - 3:30pm in Olin 375. These times are reserved just for answering your questions about the material in the course. You do not need to make an appointment in advance to attend. Simply come to my office and ask questions! You should also feel free to stop by if you just want to chat or say hello! If your schedule conflicts with my announced office hours, please contact me to set up an appointment at a different time.

Time Commitment: While the exact time commitment for the class will vary individually and over the course of the semester, I recommend that you budget approximately three out-of-class hours for every class hour to complete the assignments. I have designed the class so that it should be feasible to satisfactorily complete the requirements with approximately twelve hours per week of time commitment. If you are spending more time than this on a regular basis, I encourage you to check in with me.

Participation: Many course meetings involve in-class assignments that allow both the student and instructor to gauge student understanding. Although this work will rarely be marked for a grade, your participation is necessary both for your individual success and the success of the course overall.

Excused absences, when communicated to the instructor in a timely manner, will be accommodated to the best of the instructor's ability. However, any student who is repeatedly absent without excuse will be required to withdraw from the course. **Please contact the instructor as soon as possible** if you know you will miss class or if you miss class unexpectedly. At the sole discretion of the instructor, situations such as illness or family emergency may be considered excused absences.

Grading and Assessments

Countdown to Calculus (C2C): C2C is our pre-calculus prep program. Comprised of 19 pre-calculus modules, the program includes skeleton notes, WeBWork problem sets, WeBWork quizzes, and study guides. For each module, you are required to

- complete at least 50% of each WeBWork problem set (unlimited attempts – no penalty for wrong answers);
- complete 100% of each WeBWork quiz (unlimited attempts – no penalty for wrong answers);
- complete the study guide and submit it to me in class on **Friday, September 4th**.

The grade for C2C is equally split between the problem sets, quizzes, and study guide. You will receive full credit for completing 50% of each problem set (**not** a 50% average across all of the problems), 100% of each quiz, and the study guide. All modules are due on September 4th at 5pm, except for the study guide, which is **due in class on Friday, September 4th**.

Homework Assignments: Homework will be assigned weekly. Submissions should be neat and clear (see “Writing Guidelines” below). Your final submission should be stapled and include worked solutions for all assigned problems. Late homework will not be accepted.

A total of twelve homework assignments will be given over the course of the semester. Your three lowest homework grades will be dropped; your nine highest homework grades will be used to compute your final homework grade. Individual homework problems will be graded on a 0 to 5 scale. Feedback will be provided on your homework to indicate what is expected on exams. If you have questions, come to office hours or schedule an appointment.

You are welcome and encouraged to collaborate with others on the assignments, but you must turn in an independent write-up of your solutions, written using your own words. Students should not work together on the actual write-up of homework. Further, if you do collaborate with others, please note who you worked with at the top of the assignment.

Writing Guidelines: Below are a few guidelines for writing mathematics that you should follow when submitting work.

1. Know your audience. You should write at the level of understanding of your peers. You should provide sufficient justification so that they do not have to put extra effort into understanding a certain statement. You should also not overburden them with too many unnecessary details.
2. Justify your work. You should explain why you are using a specific formula. Your computations should be clear—the reader should be able to follow the steps of your computations with enough justification to move from one step to the next.
3. Use mathematical notation properly. Learn the correct and exact mathematical words for the concepts we work on.
4. Respect the “equals” sign ($=$). Do not use the “equals” sign non-mathematically. It should be used only in equations, and only to mean “is equal to” or “is the same as”.

5. Introduce any symbol that you use. Every variable should be defined—the reader should not have to guess what x is if it was not indicated in the question.
6. Write clearly and legibly. This holds equally for computations and for explanations.
7. Use a balanced amount of words and symbols. When you are not writing a formula, try to use sentences. Do not start a sentence with a mathematical symbol. Use complete sentences with proper spelling and grammar.

Exams: There will be three midterm exams and a final exam. The dates and times of the exams are given in the following table. Please check for possible conflicts; if any should arise, please let me know **as soon as possible**, but no later than one week prior to the scheduled exam, so that we can make alternate arrangements.

Exam	Date	Time	Location
Midterm 1	Thursday, September 24th	1:00pm – 1:50pm	Olin 268
Midterm 2	Thursday, October 22nd	1:00pm – 1:50pm	Olin 268
Midterm 3	Thursday, November 19th	1:00pm – 1:50pm	Olin 268
Final Exam	Wednesday, December 16th	7:30pm – 10:30pm	RCHM 116

Overall Grade: Your final grade for this course will be based on the following components:

Countdown to Calculus (C2C)	3%
Homework	12%
Exam A	25%
Exam B	20%
Exam C	15%
Final Exam	25%

Note that while all three midterm exams will be used to compute your final course grade, they will be weighted in your favor. Your highest exam grade will be weighted at 25% (Exam A), your second-highest exam grade will be weighted at 20% (Exam B), and your lowest exam grade will be weighted at 15% (Exam C).

Academic Integrity: Be familiar with and abide by the Bucknell University policy on cheating and plagiarism, as found in [the Honor Code](#). If you have any questions, please ask me; err on the side of caution!

AI Policy: The use of generative AI programs including LLMs like ChatGPT and Claude is not allowed for any work that is submitted for a grade or completed during class. AI programs are not a replacement for human creativity, originality, or critical thinking. However, they can be useful for generating large sets of practice materials; for example, they can be used to create a study guide or practice problems in preparation for an exam. If you have any questions about whether a particular use of AI is appropriate in this course, please do not hesitate to reach out to me.

Support and Resources

Climbing and Calculus: Looking for a change of scenery? Join me for an extra, informal office hour at [Bucknell's climbing wall](#) on Wednesday evenings at a time to be determined (most likely from 5:30pm – 6:30pm)! Come by to work through calculus questions, chat about the course, or just climb and take a break from the usual office-hour setting. No climbing experience necessary!

Quantitative Support: The Teaching and Learning Center (TLC) offers Q-Study Hours in Bertrand Library, Room 339 on Mondays – Fridays from 3pm to 6pm. Q-Study Hours are designated drop-in times for students to work on their quantitative homework with a particular emphasis on providing support with prerequisite skills or content as needed. These sessions are run by Dr. Cameron Bishop, University Quantitative Support Specialist. BERT 339 is located on the third floor of the library across from the elevator. For more information, please visit [the Quantitative Support webpage](#).

Peer-led study groups: The TLC offers peer-led study groups, which meet for about 90 minutes each week and are run by students trained to facilitate mathematical discussions tailored to the course. Registration begins at noon on Tuesday, August 25, and groups begin meeting on Sunday, September 6. More information can be found on [the TLC study group webpage](#).

TLC Tutoring: The TLC offers tutoring services for any student enrolled in our course. Students may schedule up to two 45-minute appointments each week with a peer tutor. Appointments will be visible to book by Friday, August 28. You can make an appointment or find more information on [the TLC tutoring webpage](#).

Math Department Help Sessions: The department offers drop-in help sessions that are specifically designed for students in the calculus sequence. These sessions are run by upper-class undergraduate students trained by the TLC. The sessions are held in Olin 475 on Sundays – Thursdays from 7pm – 9pm.

Academic Coaching: Coaching is open to any student who would specifically benefit from support in their development and practice of efficient learning strategies and study habits. Coaching helps build skills including time management and prioritizing goals, reading and note-taking strategies, test preparation strategies, procrastination and motivation, test anxiety, and more. Each conversation is confidential. Any student can book an appointment at <https://learningbucknell.youcanbook.me>.

Accommodation Statement: I have tried to design this course with digital accessibility and universal design for learning principles in mind, to support accessibility for all. Any student who may need an accommodation based on the impact of a disability should let me know as soon as possible and also contact the Office of Accessibility Resources (OAR) at 570-577-1188 or OAR@bucknell.edu. This office helps coordinate reasonable accommodations for those students with documented disabilities. Please visit the [OAR website](#) for more information.

Religious/Spiritual Observance: Bucknell University recognizes the value of religious practice and seeks to accommodate students' commitment to their religious and spiritual traditions whenever possible. When conflicts between holy days or other religious practice and academic scheduling arise, the instructor will make every effort to allow students to adhere to their tradition, including—when possible—excusing class absences and allowing make-up work.

If you anticipate the need to miss a class for religious reasons, you should notify me as early in the semester as possible—understanding that last-minute requests may not be able to be accommodated. A form for notification is available through the Office of Religious & Spiritual Life if you wish to use it. Once I am notified, we can have further conversation to shape a final agreement. If you have any questions/concerns or need support, you can contact the Office of Religious & Spiritual Life at Chaplain@bucknell.edu.

Classroom Behavior: Students and faculty each have a responsibility for maintaining an appropriate learning environment. In keeping with Bucknell's mission, this class strives to be an inclusive learning community, respecting those of differing backgrounds and beliefs. Professional courtesy and sensitivity are especially important with respect to

individuals and topics dealing with differences of race, color, culture, religion, creed, politics, veteran's status, sexual orientation, gender, gender identity, gender expression, age, ability, and nationality.

Class rosters are provided to the instructor with the student's legal name. I will gladly honor your request to address you by an alternate name or gender pronoun. Please advise me of this preference early in the semester so that I may make appropriate changes to my records. For more information, see the policies on [student conduct](#).

Academic Flexibility: In the event that you will need to miss class or have missed class due to health or extenuating personal matters, you need to complete the [online report form](#) for review by the Dean of Students Office (DOS). They will review your documentation and contact you with any questions or to request further information. If the issue meets the criteria for an absence notification or academic flexibility request, DOS will provide you with a letter to provide to me as your instructor. Please note that such a letter is a notification/request only; it is in my purview as the instructor to determine whether an absence is excused and/or how flexibility can be extended in the context of this course.

Discrimination and Harassment: Bucknell University does not discriminate on the basis of race, color, national origin, sex, disability, religion, age, veteran status, gender identity, marital status, sexual orientation, gender expression, or any characteristic protected by law in its educational programs and activities, admissions or employment.

As part of the Bucknell community, I value and want to engage with varied perspectives and experiences. I will work to ensure that everyone feels welcomed, respected, and safe. I expect others to do the same. Please alert me if any aspect of this course raises concerns about equity or inclusion for you. The University's [Division of Equity and Inclusive Excellence website](#) also has a variety of resources should you need them.

Honor Code: Students who violate Bucknell University standards of academic integrity are subject to disciplinary sanctions, including failure in the course and suspension from the University. Since dishonesty in any form harms the individual, other students, and the University, policies on academic integrity are strictly enforced. Examples of academic dishonesty include, but are not limited to: (1) copying from another student's homework, quiz, or exam; (2) allowing another student to copy your work; (3) copying homework solutions from the text solutions manual; and (4) copying from journal articles, books, or online resources. With proper citation, some of these examples would not be considered academic dishonesty. You should familiarize yourself with the academic integrity guidelines found in the current [student handbook](#).

Wellness Statement: In this classroom and on Bucknell's campus at large, we support mental health efforts. Any student who is struggling and believes this may impact their performance in the course is encouraged to contact their Associate Academic Dean or the Dean of Students at 570-577-1601 for support. Furthermore, please approach me if you are comfortable in doing so. This will enable me to provide resources and support. If immediate mental health assistance is needed, call the [Bucknell Student Counseling Center](#) at 570-577-1604 (24/7).

A Final Note: I, like you, am human. If something comes up and you need extra flexibility with this course, please reach out to me. The sooner you let me know that something is going on, the sooner we can work together to figure out reasonable accommodations.